

Lightwave Systems: Design and Performance

Exploring OptiSystem' Samples

Agrawal, Chap. 5

Computer Aided Design

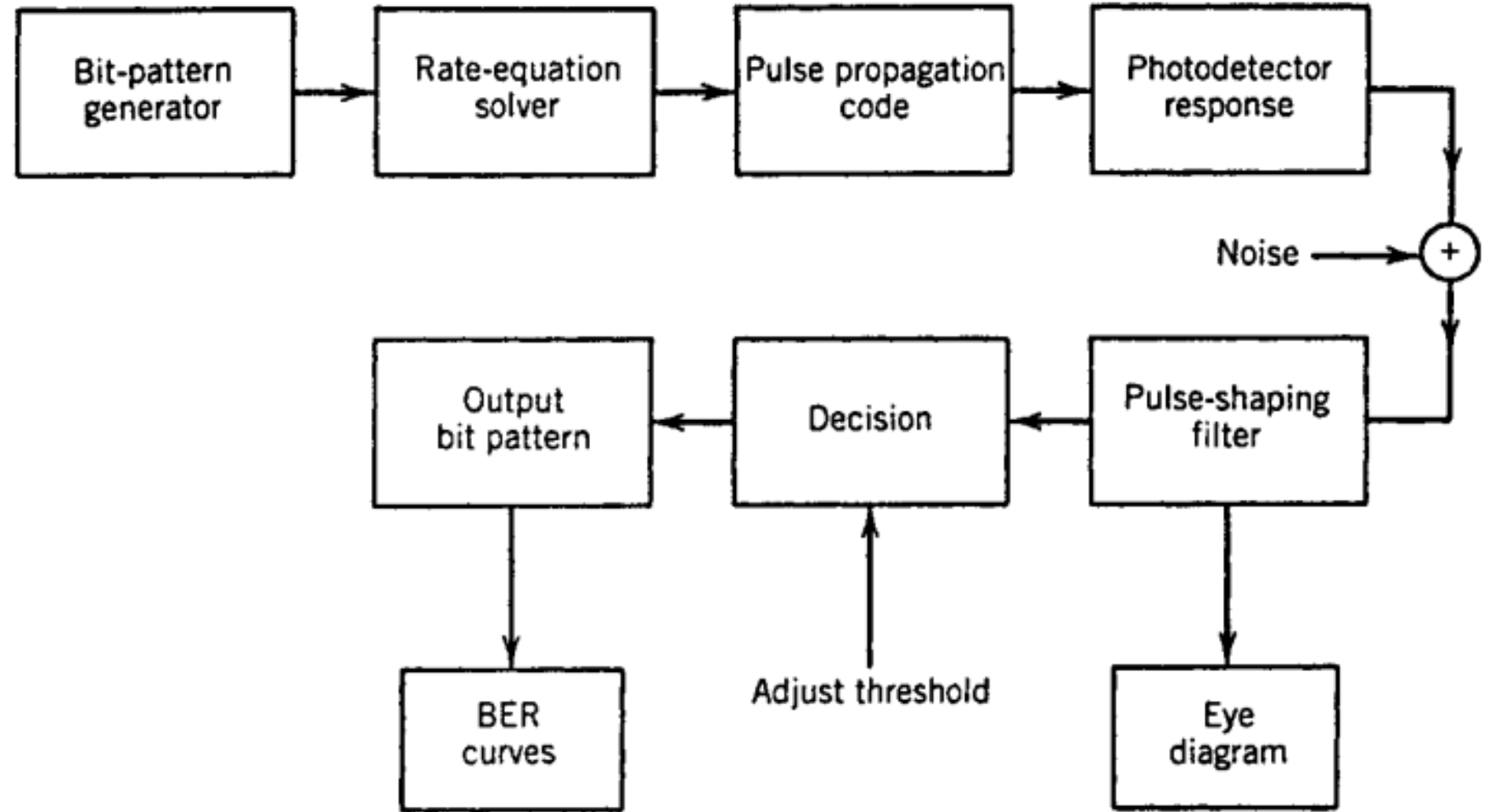


Figure 5.15: Steps involved in computer modeling of fiber-optic communication systems.

Theoretical
approach and
illustrative
samples

OptiSystem

Tutorials - Volume 1

Optical Communication System Design Software

OptiSystem

Tutorials - Volume 2

Optical Communication System Design Software

System Performance

Optical receivers

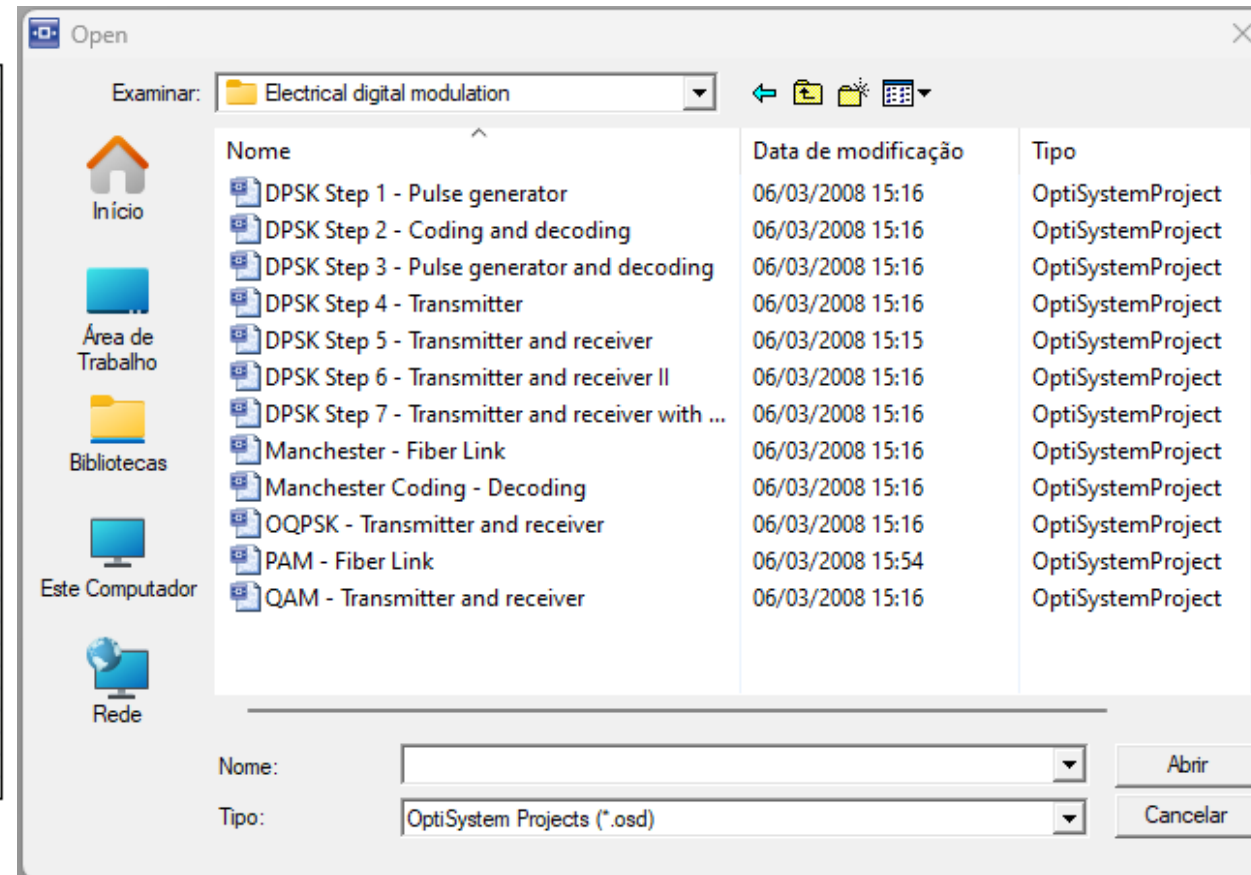
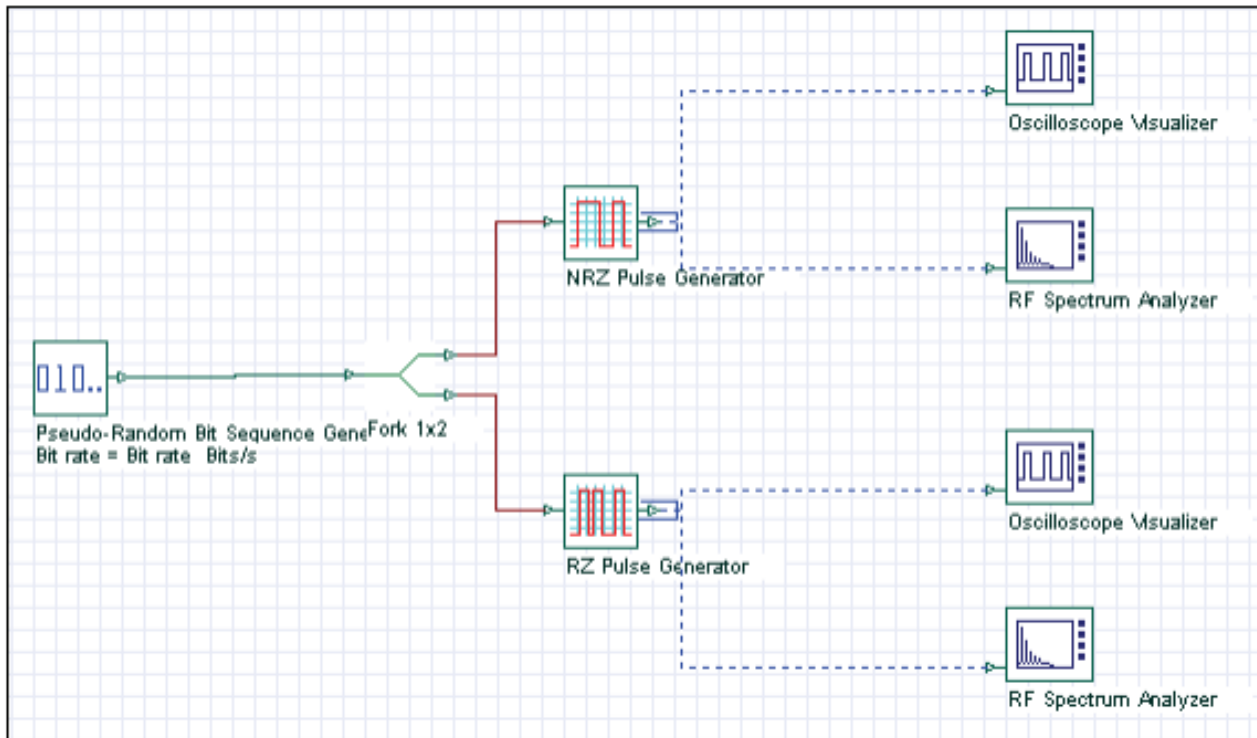
The first portion of this section provides introductory information about [Modulation formats](#).

The remainder of the section contains the following advanced and illustrative [simulation projects](#):

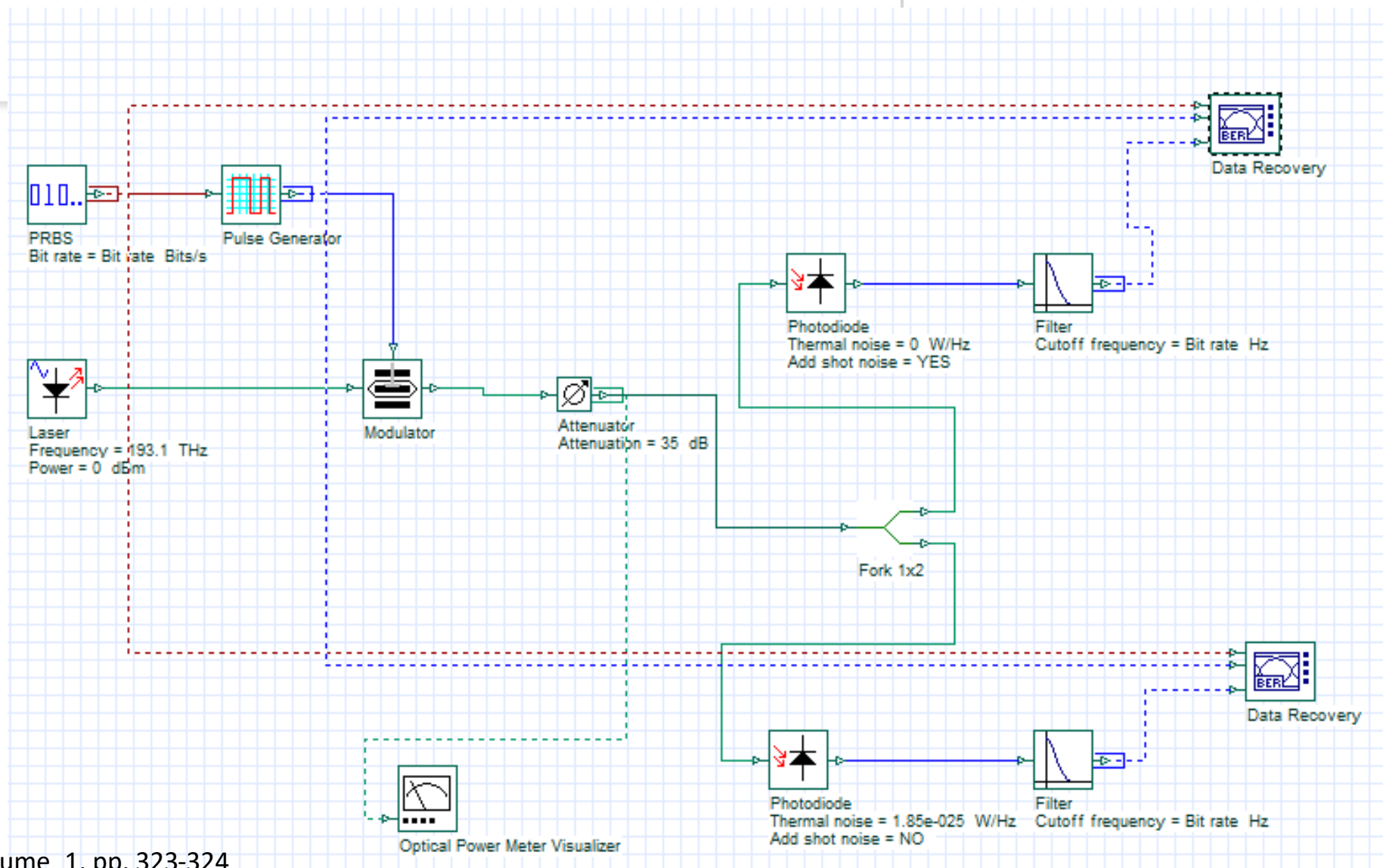
- [Extracting the thermal noise parameter for a specific receiver sensitivity](#)
- [Receiver noise—PIN](#)
- [Receiver noise—Shot noise enhancement with APD](#)
- [Receiver sensitivity—Bit error rate \(BER\)](#)
- [Receiver sensitivity—Minimum input power](#)
- [Sensitivity degradation—Extinction ratio](#)
- [Signal degradation—Jitter](#)

Modulation Formats

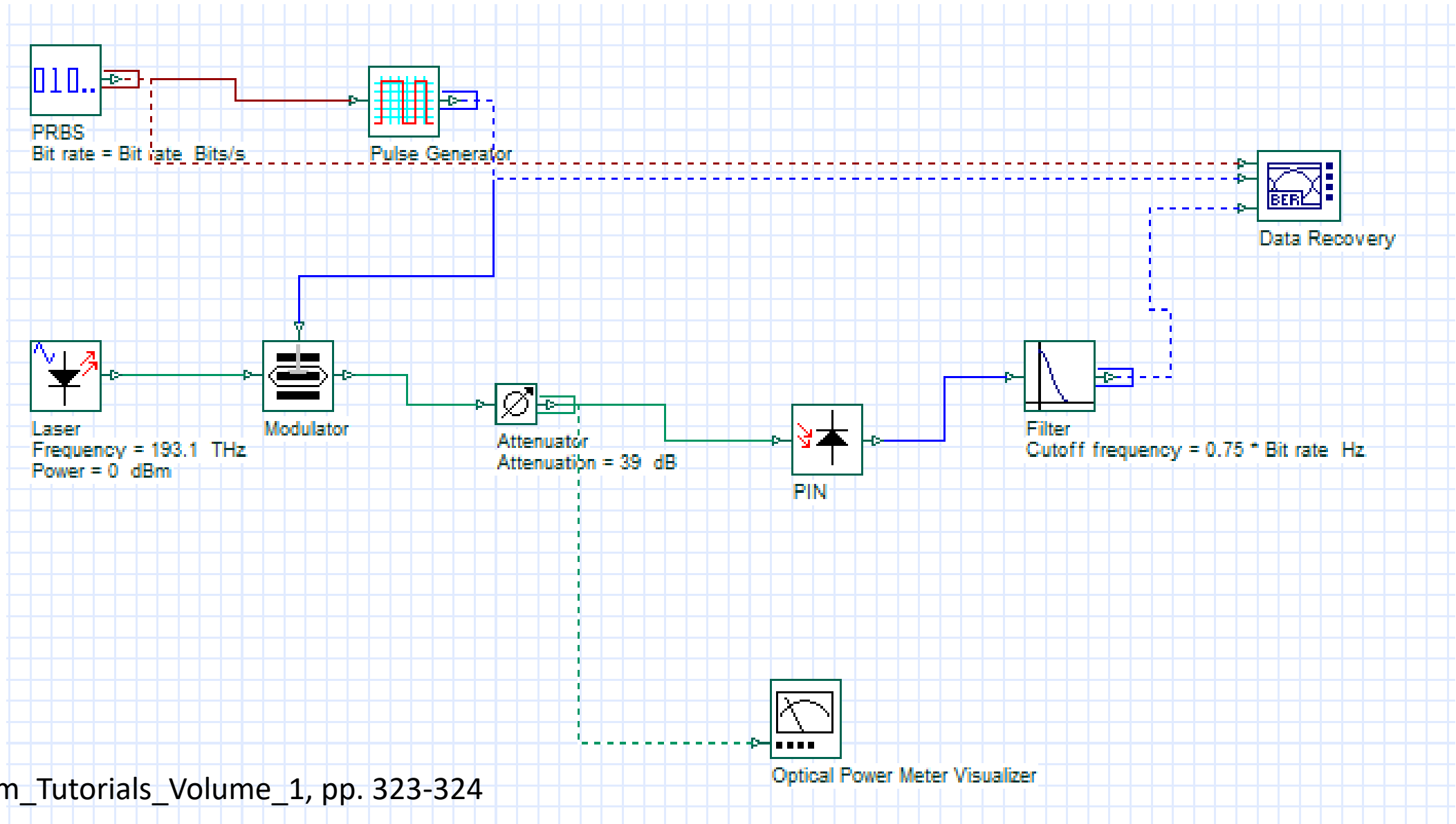
Figure 1 Modulation Formats



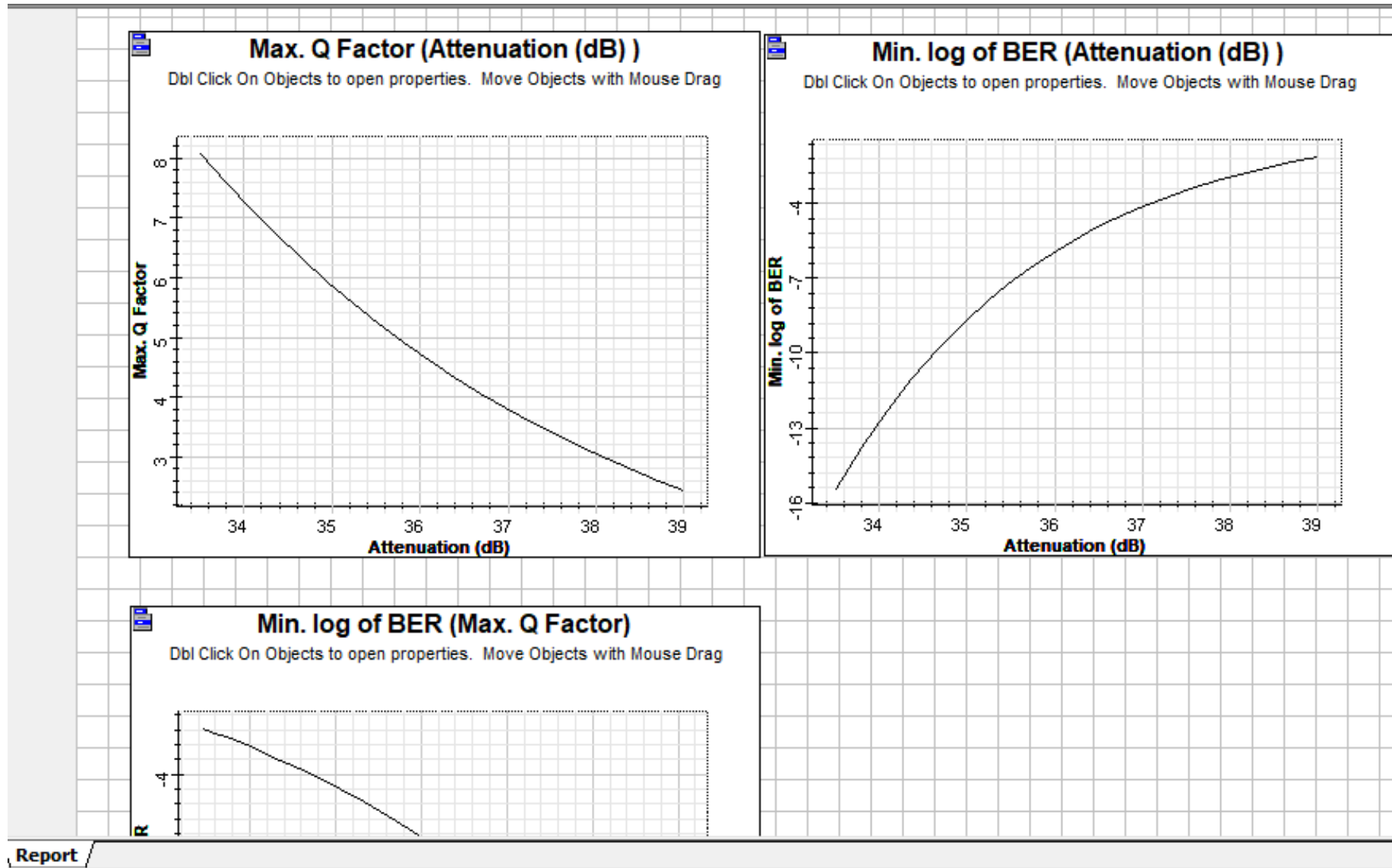
Receiver Noise - PIN



Receiver sensitivity—Bit error rate (BER) and Q factor



Receiver sensitivity—Bit error rate (BER) and Q factor



Dispersion Problem and Its Solution

- Systems built during 1980s used standard fibers with their zero-dispersion wavelength near $1.3 \mu\text{m}$.
- Standard fibers have large dispersion near $1.55 \mu\text{m}$.
- Operation near zero-dispersion wavelength not realistic for WDM systems.
- Even with DFB lasers, transmission distance is limited to

$$L < \frac{1}{16|\beta_2|B^2} = \frac{\pi c}{8\lambda^2|D|B^2}.$$

- $L < 35 \text{ km}$ at $B = 10 \text{ Gb/s}$ if we use $|\beta_2| \approx 21 \text{ ps}^2/\text{km}$.
- Dispersion must be compensated or managed using a suitable technique before old systems can be upgraded to 10 Gb/s .

Engineering the fiber nonlinearities and dispersion

Layout: Version 2

Author: Optiwave

Monday, February 19, 2001

Sweep Iteration: 3/3

